2.2 Group 2

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.2 Group 2
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Samples of barium and calcium are added separately to beakers of cold water which contain a few drops of litmus solution.

Which observations will be made with **only** the calcium and **not** with the barium?

1 A gas is evolved

2 A white suspension appears in the water

3 The solution turns blue

A. 1 only

B. 2 only

C. 1 and 2

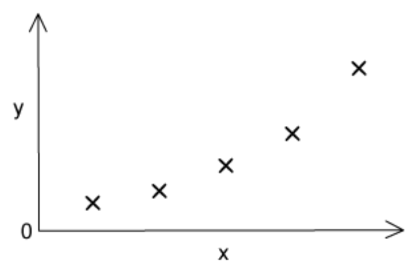
D. 1, 2 and 3

[1 mark]

Question 2

The sketch graph shows the variation of one physical or chemical property with another for the Group II elements.

Use of the periodic table is relevant to this question.



What are the correct labels for the axes?

	***X*-axis**	***Y*-axis**
A	First ionisation energy	Atomic number
B	First ionisation energy	Atomic radius
C	Atomic number	Melting point
D	Atomic number	Mass number

[1 mark]

Question 3

X is a Group II metal. It forms a hydroxide which is more soluble than calcium hydroxide. It forms a sulfate which is more soluble than barium sulfate.

What could be the identity of **X**?

1 Strontium

2 Beryllium

3 Magnesium

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 4

The group I metals have lower melting points than the Group II metals.

Which factors result in Group II metals having higher melting points?

1 Group II metals have higher first ionisation energy than the corresponding group I metal.

2 There are smaller interatomic distances in the metallic lattices of the Group II metals.

3 More electrons are available from each Group II metal atom for bonding the atom into the metallic lattice.

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 5

Pieces of pure solid magnesium and calcium are burned separately with oxygen.

Pieces of pure solid magnesium and calcium are then reacted separately with cold water.

Which row correctly describes the observations from these reactions?

	Magnesium with oxygen	Calcium with oxygen	Magnesium with cold water	Calcium with cold water
A	white flame	red flame	rapid bubbling	slow bubbling
B	red flame	white flame	rapid bubbling	slow bubbling
C	red flame	white flame	slow bubbling	rapid bubbling
D	white flame	red flame	slow bubbling	rapid bubbling

[1 mark]

Question 6

What happens when a piece of strontium is placed in cold water?

A. A vigorous effervescence occurs.

B. Bubbles of gas form slowly on the strontium.

C. The strontium floats on the surface of the water and reacts quickly.

D. The strontium glows, and a white solid is produced.

[1 mark]

Question 7

When barium is burnt in oxygen, what colour is the flame?

A. Green

B. White

C. Red

D. Yellow

[1 mark]

Question 8

Which statements about calcium oxide are correct?

1 It is produced when calcium nitrate is heated.

2 It can be reduced by heating with magnesium.

3 It reacts with cold water.

A. 1 only

B. 1 and 2

C. 1 and 3

D. 2 and 3

[1 mark]

Question 9

The oxides BaO, CaO, MgO and SrO all produce alkaline solutions when added to water.

Which oxide produces a saturated solution with the highest pH?

A. BaO (aq)

B. CaO (aq)

C. MgO (aq)

D. SrO (aq)

[1 mark]

Question 10

Rat poison needs to be insoluble in rainwater but soluble at the low pH of stomach contents.

What is a suitable barium compound to use for rat poison?

A. Barium carbonate

B. Barium chloride

C. Barium hydroxide

D. Barium sulfate

[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Group II elements get more reactive as you go down the group.
- Both calcium and barium will react with water to form hydrogen gas, barium will react more readily.
- When calcium reacts with water, a suspension of calcium hydroxide is formed.
- The solubility of the hydroxides increases as you go down the group so barium hydroxide would not produce a white suspension.

A & **C** are incorrect as both reactions would release hydrogen gas.

D is incorrect as both reactions would produce an alkaline solution and turn litmus solution blue.

2

The correct answer is **D** because:

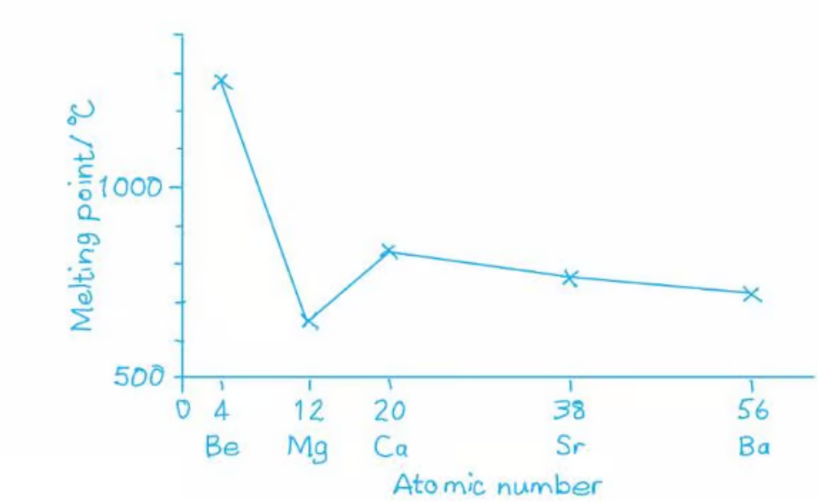
- As the atomic number increases as you go down Group II so does the mass.

Group II element	Atomic number	Atomic mass
Beryllium (Be)	4	9
Magnesium (Mg)	12	24.3
Calcium (Ca)	20	40.1
Strontium (Sr)	38	87.6
Barium (Ba)	56	137.3

- This is the same pattern that is seen in the graph.

A & **B** are incorrect as the first ionisation energy would be on the y-axis when compared with the atomic number or atomic radius

C is incorrect as A graph showing melting points for Group II would produce a graph that looks as follows:



3

The correct answer is **A** because:

- The solubility of the hydroxides increases as you go down the group.
- The Group II sulfates get less soluble going down the group.
- The identity of X can only be strontium.

4

The correct answer is **C** because:

- The metallic radius is half the distance between the nuclei in a giant metallic lattice. This is smaller as we go across the period.
  - The metallic radius is generally smaller in Group II metals than group 1.
- The number of protons in the nucleus increases, holding the electrons in the same shell more tightly, and the atoms become smaller. Resulting in higher melting

points as the atoms are closer together.

- Group II metals have 2 valence electrons so one more electron per atom available for bonding in the metallic lattice.

Statement 1 is incorrect because the first ionisation energy is the energy needed to remove the most loosely held electron from each of one mole of gaseous atoms to make one mole of singly charged gaseous ions. The melting point of the metal would not be affected by the ionisation energy.

5

The correct answer is **D** because:

- Magnesium burns with a bright white flame.
- Calcium burns with a brick-red colour flame.
- Group II elements get more reactive as you go down the group.
- Calcium will react more readily with cold water forming hydrogen gas and a suspension of calcium hydroxide.
- So calcium will show rapid bubbling with cold water and magnesium slow bubbling.

6

The correct answer is **A** because:

- Group II elements get more reactive as you go down the group.
- Strontium will react more readily with cold water forming hydrogen gas and a soluble solution of strontium hydroxide

- Strontium will react more readily with cold water forming hydrogen gas and a soluble solution of strontium hydroxide.
- Strontium will show rapid effervescence as it reacts with cold water.

B is incorrect as strontium will produce bubbles rapidly.

C is incorrect as strontium will not float on the surface of the water due to higher densities.

D is incorrect as when placed in cold water strontium does not glow white, and strontium hydroxide is soluble.

7

The correct answer is **A** because:

- The flame test for barium shows that it burns with an apple-green colour flame.

B is incorrect as magnesium burns with a white light.

C is incorrect as calcium gives a brick-red colour and strontium a scarlet/red colour.

D is incorrect as sodium burns with a yellow flame.

8

The correct answer is **C** because:

- When heated calcium nitrate will thermally decompose to calcium oxide and nitrogen dioxide.
  - $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$
- Calcium oxide reacts with water to form calcium hydroxide.
  - $CaO (s) + H_2O (l) \rightarrow Ca(OH)_2 (s)$

Statement 2 is incorrect as Group II elements get more reactive as you go down the group. Therefore, calcium is more reactive than magnesium and will not be reduced by it.

9

The correct answer is **A** because:

- Group II elements get more reactive as you go down the group.
- As you go down Group II the solubility of the metal oxides increases and the more alkaline the solutions.
- Barium is the lowest Group II element in the list so will produce a saturated solution with a high pH.

10

The correct answer is **A** because:

- The solubility of the barium compounds in water and acid are shown in the table below.

Compound	**Soluble in water**	**Soluble in stomach acid**
Barium carbonate	No	Yes
Barium chloride	Yes	No
Barium hydroxide	Yes	Yes
Barium sulfate	No	No

- The only option from the table would be barium carbonate as it is not soluble in water and soluble in stomach acid.